

S. A. I. Shouborno

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WORK EXPERIENCE

Cyber-Physical Systems Intern, Robert Bosch LLC, Pittsburgh

May 2025 - Aug 2025

- Developed a contrastive multimodal representation learning model connecting vision, sensor, program traces, and texts, and performed language-guided few-shot functional state classification

Research and Teaching Assistant, Worcester Polytechnic Institute

Aug 2023 - Present

- TA for Electrical & Computer Eng. courses, including the graduate course On-Device Deep Learning
- Summer RA for Bringing Awareness through Systems for Humans (BASH) Lab

Machine Learning Engineer, Hishab Technologies Ltd., Bangladesh

Feb 2021 - Jul 2023

- Projects involved Natural Language Processing (Conversational AI, ASR, NER, etc.).
- Worked with Hishab's partner Chowa Giken (Japan) on a Computer Vision project for 1 month.

EDUCATION

PhD in Electrical and Computer Engineering

Aug 2023 - Present

Worcester Polytechnic Institute, USA

MSc in Electrical and Computer Engineering

Aug 2023 - Dec 2025

Worcester Polytechnic Institute, USA

Thesis: Interpretable and Efficient AI for Sensor-rich Mobile and Embedded Systems

Supervisor: Dr. Bashima Islam

Courses: Deep Learning (Fabricio Murai), Natural Language Processing (Xiaozhong Liu),
Digital Image Processing (Ziming Zhang), Wearable/Mobile Sensors & Systems (Ted Clancy)

CGPA: 4.00/4.00 | wpi.edu

BSc in Electrical and Electronic Engineering

Feb 2016 - Feb 2021

Bangladesh University of Engineering and Technology, Bangladesh

Thesis: Audiovisual emotion recognition using multi-head attention based neural network

Supervisor: Dr. Celia Shahnaz

CGPA: 3.40/4.00 | buuet.ac.bd

PATENT AND PUBLICATIONS

- AI MUNTASIR, T., ISLAM, M. T., & **IMRAN, S. A.** (2025). System and method for optimizing a user interaction session within an interactive voice response system. U.S. Patent Application No. 18/378,118.
- Asif Imran Shouborno, S.**, Khan, M. N. H., Biswas, S., & Islam, B. (2025, October). LLaSA: A Sensor-Aware LLM for Natural Language Reasoning of Human Activity from IMU Data. In Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (pp. 893-899).
- Mahmud, T. I., **Imran, S. A.**, & Shahnaz, C. (2022). Res-SE-ConvNet: A deep neural network for hypoxemia severity prediction for hospital in-patients using photoplethysmograph signal. IEEE Journal of Translational Engineering in Health and Medicine, 10, 1-9.
- Imran, S. A.**, Islam, M. T., Shahnaz, C., Islam, M. T., Imam, O. T., & Haque, M. (2020, December). COVID-19 mRNA vaccine degradation prediction using regularized LSTM model. In 2020 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE) (pp. 328-331).
- Imam, O. T., Haque, M., Shahnaz, C., **Imran, S. A.**, Islam, M. T., & Islam, M. T. (2020, December). Detection of tuberculosis from chest X-ray images based on modified inception deep neural network model. In 2020 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE) (pp. 360-363).
- Qayyum, A.B.A., Hassan, K.N., Anika, A., Shadiq, M.F., Rahman, M.M., Islam, M.T., **Imran, S.A.**, Hossain, S. and Haque, M.A. (2020). DOANet: a deep dilated convolutional neural network approach for search and rescue with drone-embedded sound source localization. EURASIP Journal on Audio, Speech, and Music Processing, 2020(1), p.16.

- Islam, M. T., **Imran, S. A.**, Arefeen, A., Hasan, M., & Shahnaz, C. (2019, November). Source and camera independent ophthalmic disease recognition from fundus image using neural network. In 2019 IEEE International Conference on Signal Processing, Information, Communication & Systems (SPICSCON) (pp. 59-63).
- Qayyum, A.B.A.A., Anika, A., Miah, M.M.M., Rahman, M.M., Hasan, K.N., Islam, M.T., **Shouborno, S.A.I.**, Shadiq, M.F. and Haque, M.A. (2019, November). Direction of Arrival Estimation through Noise Suppression: A Novel Approach using GSC Beamforming and Room Acoustic Simulation. In 2019 IEEE International Conference on Signal Processing, Information, Communication & Systems (SPICSCON) (pp. 104-108).
- **Shouborno, S.A.I.**, Mahmud, T.I., Ishraq, N., Ali, R., Joy, T.H., Fattah, S.A., Bhattacharjee, A. and Shahnaz, C. (2019, November). Complete Automation of an E-commerce System with Internet of Things. In 2019 IEEE International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON) (pp. 81-86).

HONOURS AND AWARDS

5th at IEEE VIP Cup 2020 – IEEE ICIP 2020, UAE

Jul 2020 - Sep 2020

Challenge: Real-time vehicle detection and tracking at junction using a fisheye camera

4th at IEEE SP Cup 2020 – IEEE ICASSP 2020, Spain

Nov 2019 - Mar 2020

Challenge: Unsupervised abnormality detection using intelligent & heterogeneous autonomous systems

2nd at IEEE VIP Cup 2019 – IEEE ICIP 2019, Taiwan

Apr 2019 - Sep 2019

Challenge: Activity Recognition from Body Cameras

VOLUNTEERING EXPERIENCE

Chairperson, IEEE Signal Processing Society BUET SB Chapter

Jul 2019 - Mar 2021

- Arranged IEEE SPS Winter School 2019 on Multimodal Signal Processing
- Helped the chapter become the largest IEEE Student Branch Chapter of Bangladesh and receive IEEE SPS Student Branch Chapter Growth Reward

Ambassador (2017-2019) and Country Lead (2019), IEEEExtreme Programming Contest

- Arranged programming workshops
- Participated in 2016, 2017 & 2018 (National champion in 2017 & 2018; 22nd worldwide in 2018)

SKILLS

Machine Learning (Python, Tensorflow, Keras, PyTorch, MATLAB, Computer Vision, NLP, MLOps)

Signal Processing (Image, Video, and Audio Processing; Biomedical Signal Processing)

Competitive Programming, LaTeX, Git, Microcontrollers, FPGA, IoT, Web Development, Project Management

PROJECTS

- **Interpretable Physics-Informed Neural Network for Cuffless Blood Pressure Prediction, 2026**

In an ongoing research collaboration with Kireevlab (UMass Amherst), we are building methods that can accurately estimate blood pressure from graphene electronic tattoo patches for easy and continuous health monitoring.

- **Agentic AI for Physics-aware Code Execution Debugging for Industrial Cyber-Physical Systems, 2026**

In an ongoing research collaboration with Bosch, we are building a code execution trace generation and debugging AI to improve diagnostics and to help with faster deployment for industrial machine operators.

- **LLM as a Discriminator for Evaluating Machine-generated News Headlines, 2024**
- **Large Language and Vision Assistant with Reasoning for Distinguishing Real-Fake Images, 2024**
- **Zero-glance Single-pass Machine Unlearning for Text Classification, 2023**
- **Deep Unlearning with Explainable AI and Saliency Maps, 2023**
- **Telephony conversational AI solutions for bank, MFI, and MSME financial services, 2021-2023**
- **Car interior image improvement with image processing and machine learning, 2021**
- **Real-time distortion classification in laparoscopic videos, 2020**